

Quiz (Habanero)

- Q1.** If the temperature of a city is expected to increase by a factor of 2^3 over the next 5 years, and it is currently 15^{circ}C , what will the temperature be in 5 years?
- (A) 30^{circ}C
(B) 40^{circ}C
(C) 60^{circ}C
(D) 80^{circ}C
(E) $120^{\text{circ}}\text{C}$
- Q2.** A certain type of plant can grow at a rate of 3^2 times its original size when exposed to optimal rainfall conditions. If the plant is initially 5cm tall, how tall will it be after optimal rainfall?
- (A) 5 cm
(B) 15 cm
(C) 30 cm
(D) 45 cm
(E) 60 cm
- Q3.** Simplify the expression $x^2 \cdot x^4$
- (A) x^6
(B) x^8
(C) x^{10}
(D) x^{12}
(E) x^{14}
- Q4.** A researcher measures the growth of a particular species of ecosystem, finding that its population grows by a factor of $(2^2)^3$ per year. What is the annual growth factor?
- (A) 2^6
(B) 2^8
(C) 2^{10}
(D) 2^{12}
(E) 2^{14}
- Q5.** The volume of a cloud can be calculated using the formula $V = l^3$, where l is the length of the cloud in kilometers. If the length is 2^2 km, what is the volume?
- (A) 8 km^3
(B) 16 km^3
(C) 32 km^3
(D) 64 km^3
(E) 128 km^3
- Q6.** The rate of rainfall in a certain region can be modeled by the equation $R = 3^t$, where R is the rate of rainfall and t is time in hours. What is the rate of rainfall after 2 hours?
- (A) 3 mm/h
(B) 9 mm/h
(C) 18 mm/h
(D) 27 mm/h
(E) 81 mm/h
- Q7.** A group of scientists study the effects of climate change, finding that the level of greenhouse gases in the atmosphere increases by a factor of 2^3 every decade. What is the increase after 3 decades?
- (A) 2^6
(B) 2^9
(C) 2^{12}
(D) 2^{15}
(E) 2^{18}
- Q8.** The amount of solar radiation that hits the Earth's surface can be calculated using the formula $S = 4^t$, where S is the amount of solar radiation and t is time in hours. What is the amount of solar radiation after 1 hour?
- (A) 4 J/m^2
(B) 8 J/m^2
(C) 16 J/m^2
(D) 32 J/m^2
(E) 64 J/m^2

Open Ended Questions (FRQ)

- Q9.** A certain type of ecosystem is sensitive to temperature changes, with its population decreasing by a factor of 3^2 for every 2^{circ}C increase in temperature. If the temperature is expected to increase by 2^3 degrees over the next 10 years, and the current population is 1000, what will the population be in 10 years? Show your work and explain your reasoning.
- Q10.** The level of oxygen in a lake can be modeled by the equation $O = 2^{t-2}$, where O is the level of oxygen and t is time in months. If the current level of oxygen is 16, what will the level be after 5 months? Show your work and explain your reasoning.

Chef's Tips

- When evaluating expressions with exponents, simplify the exponents first before performing any other operations
- Use the order of operations (PEMDAS) when simplifying expressions with exponents
- Make sure to consider the units when working with real-world applications of exponents